1. SYSTEM IDENTIFICATION

1.1. System Name/Title: State Institute of Science and Technology (SIST) Web Infrastructure System (SWI)

1.1.1. System Categorization:
Confidentiality: Moderate
Integrity: Moderate
Availability: Moderate
Overall Security Categorization: Moderate Impact System

1.1.2. System Unique Identifier: SIST-SWI-AWS-001

1.2. General Description/Purpose of System: What is the function/purpose of the system?

The SIST Web Infrastructure System provides the cloud based infrastructure that supports several vital organizational applications. These applications include the Financial and Contract Information Network (FCIN), the Graduate Management Admission Testing application, and the Human Resources system. Web servers, and database servers are on Amazon EC2 instances make up the multi-tier architecture of the system, which runs in an Amazon Web Services environment. The infrastructure guarantees the availability, confidentiality, and integrity of sensitive government data while allowing users to safely access organizational services. Audit logging systems have been put in place using AWS CloudTrail, Amazon CloudWatch Logs, and Amazon S3.

1.2.1. Number of end users and privileged users:
Total users : 470
Privileged users: 20

Roles of Users and Number of Each Type:

Roles	Number of Users	Number of Administrators/ Privileged Users
Students	150	0
System administrators	5	5
Database administrators	3	3
Application administrators	10	10
Security administrators	2	2
General employees	300	0

1.3. General Description of Information: CUI information types processed, stored, or transmitted by the system are determined and documented. For more information, see the CUI Registry at https://www.archives.gov/cui/registry/category-list.

The system processes and stores CUI including:

- Personal Information

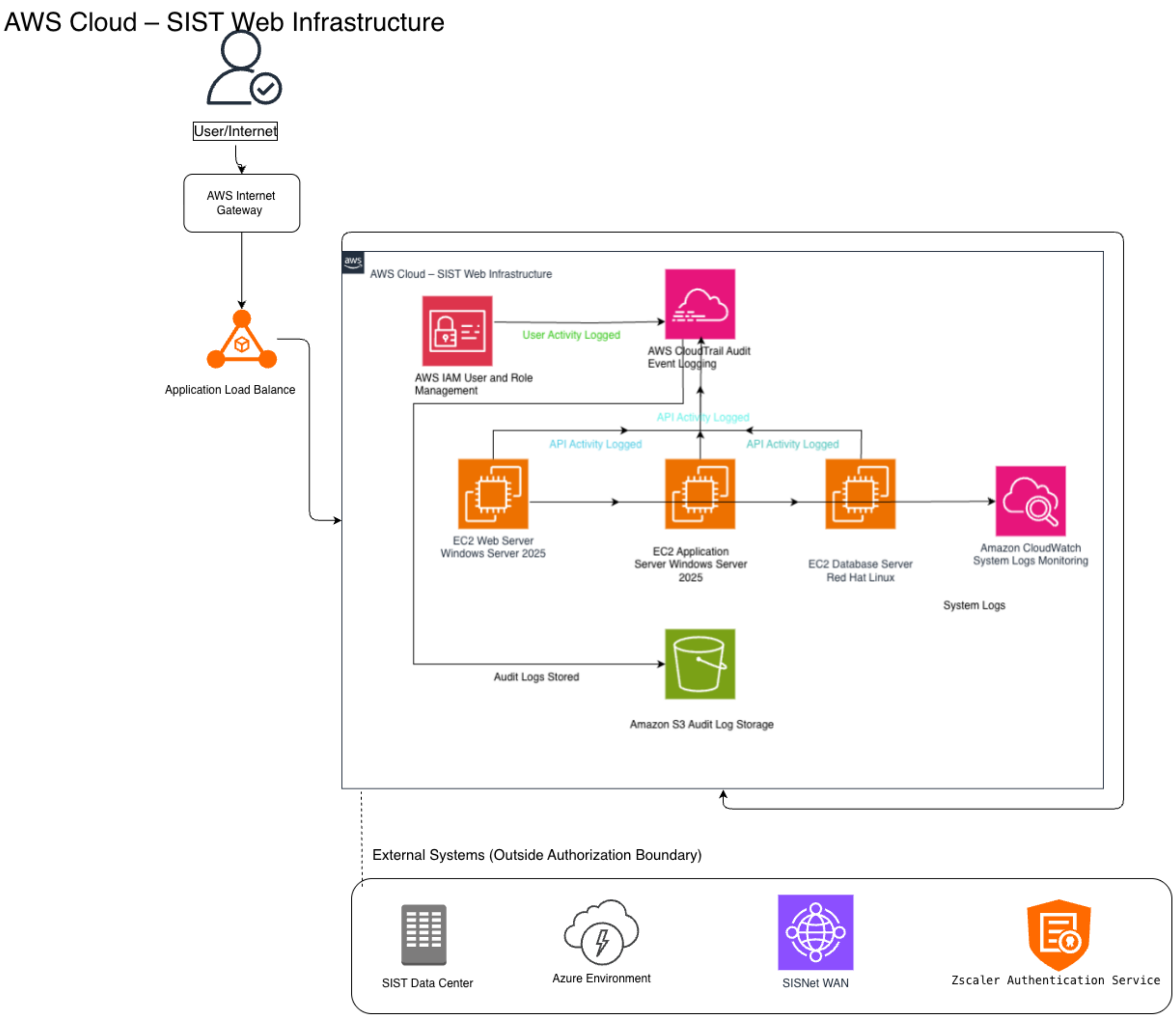
- Student records
- Financial and contract data
- System audit logs and security monitoring information

2. SYSTEM ENVIRONMENT

Include a detailed topology narrative and graphic that clearly depicts the system boundaries, system interconnections, and key devices. (Note**:** *this does not require depicting every workstation or desktop*, but include an instance for each operating system in use, an instance for portable components (if applicable), all virtual and physical servers (e.g., file, print, web, database, application), as well as any networked workstations (e.g., Unix, Windows, Mac, Linux), firewalls, routers, switches, copiers, printers, lab equipment, handhelds). If components of other systems that interconnect/interface with this system need to be shown on the diagram, denote the system boundaries by referencing the security plans or names and owners of the other system(s) in the diagram.

The SIST Web Infrastructure System runs inside an AWS Virtual Private Cloud (VPC). Users connect securely using HTTPS over the internet. Incoming traffic goes through an Internet Gateway and reaches the web server on Amazon EC2 instance. The web server handles user's requests and sends application logic requests to application server than talks to a database server running Red Hat Enterprise Linux, which stores all application data and records.

AWS native services are used to implement audit logging and monitoring. While Amazon CloudWatch Logs gathers operating system and application logs from the EC2 instances, AWS CloudTrail documents API activity and administrative operations throughout the AWS ecosystem. These logs are safely kept in Amazon S3 for long term -reservation and forensic investigation. Additionally, the system establishes connections with external services that are located outside of the system authorization boundary, including the SIST data center, Azure environment, SISTNet WAN, and Zscaler authentication services.



https://app.diagrams.net/

2.1. Include or reference a **complete and accurate** listing of all hardware (a reference to the organizational component inventory database is acceptable) and software (system software and application software) components, including make/OEM, model, version, service packs, and person or role responsible for the component.

- Hardware resources are provided through AWS. A complete hardware inventory is maintained in the SIST configuration management database.

Hardware Component	**Vendor**	**Model/Service**	**Responsible Role**
Computer Infrastructure	AWS	Amazon EC2	Cloud Administrator
Storage Infrastructure	AWS	Amazon S3	Cloud Administrator
Networking Infrastructure	AWS	AWS VPC	Network Administrator

2.2. List all software components installed on the system.

Software	Version	Purpose
Window Server	2025	Web and Application Servers
RedHat Enterprise Linux	9	Database server
Oracle Database	19c	Database management
Java Runtime environment	17	Application runtime
AWS CloudTrail	Managed service	API audit logging
Amazon CloudWatch Logs	Managed service	System monitoring
Amazon S3	Managed service	Log Storage

2.3. Hardware and Software Maintenance and Ownership - Is all hardware and software maintained and owned by the organization?

- Yes. The ISST Web Infrastructure System’s hardware and software are either supplied by Amazon Web Services through a managed cloud service agreement or are maintained by the company. AWS maintains the underlying physical infrastructure, while SIST is in charge of protecting operating systems, apps, and data within the cloud environment under AWS shared responsibility model.